

2.5 hours on examples

Problem 1 (Reim's theorem)

If $m_1 \parallel m_2$, then $l_1 \parallel l_2 \parallel l_3$ and we're done.

If $m_1 \not\parallel m_2$, we let $P = m_1 \cap m_2$. We let $A = l_2 \cap m_1$, $B = l_2 \cap m_2$, $C = l_1 \cap m_1$, $D = l_1 \cap m_2$, $E = l_3 \cap m_1$, $F = l_3 \cap m_2$. By Power of a Point on $ABCD$,

$$PA \cdot PC = PB \cdot PD$$

By Power of a Point on $EFCD$,

$$PC \cdot PE = PD \cdot PF$$

Therefore, $\frac{PA}{PE} = \frac{PB}{PF} \Leftrightarrow l_1 \parallel l_3$.

Problem 4 (APMO 2015/1)

(All $\angle$ are directed)

We prove that AB and VW subtend to the same angle, more specifically that $\angle VZW = \angle BCA$.

Claim 1: $XD \parallel AV$

Using angle chasing, we get that,

$$\angle BXD = \angle BZD = \angle BAV$$

Therefore, we have that $XD \parallel AV$.

Claim 2: $ZDCY$ is cyclic.

$$\angle ZDY = \angle ZDX = \angle ZVA = \angle ZCA = \angle ZCY$$

Therefore, $\angle BCA = \angle BZY = \angle VZW$ as desired.

Problem 6 (UK 2015)

(All $\angle$ are directed)

Claim: $CDPQ$ is cyclic.

Proof: Note that,

$$\angle PDQ = \frac{\angle ADB}{2} = \frac{\angle ACB}{2} = \angle PCQ$$

Therefore,

$$\angle QPC = \angle QDC = \angle BDC = \angle BAC$$

Therefore, $PQ \parallel AB$.

Problem 8 (HMIC 2016/2)

We let O' be the reflection of O over BC and H' be that of H . By reflecting the orthocenter, we can reflect $(ABC) = (BH'C)$ over BC to get (BHC) so O' is the center of (BHC) .

We let B_A be $AH \cap BC$, M be the midpoint of BC , and X be the center of (MHN) . By a homothety centered at H of scale $\frac{1}{2}$ on (BHC) , HO' is the diameter of (MHN) . Therefore, if O is on the circle,

$$\angle HOO' = 90 \Leftrightarrow HO \parallel BC \Leftrightarrow \angle H'HO = 90$$

X would also be the on the perpendicular bisector of OO' , which is BC . This means that H' also lies on (MHN) . We then get $\angle HH'O' = 90$ as well.

Therefore, $HOO'H'$ is a rectangle so O, X, H' are collinear and (MHN) is internally tangent to (ABC) .

Problem 10 (IMO 2004/1)

By SAS on MOR and NOR , $MR = NR$. We let the circle with radius RN centered at R intersect AC again at N' . Note that N'' the point on ray AC where $AM = AN''$ must also lie on this circle. Therefore, exactly one of N, N' must be N'' .

Case 1: $N = N''$

This means that $AM = AN$ so by power of a point on A , $AB = AC$ as well, which is not allowed.

Case 2: $N \neq N''$

This means that $N' = N''$, so $\angle AMR = \angle AN'R = 180 - \angle ANR$. Therefore, $AMNR$ is cyclic.

We let X be the point on BC such that (AMX) passes through R . By Miquel Point, (CNX) also passes through R . Therefore, (AMR) and (CNR) both pass through the point X on BC .

Problem 12 (Germany 2012/3)

We let D be the midpoint of major arc BC .

Claim: $AB + AC = AD$

Proof:

By Law of Sines,

$$AB + AC = AD$$

$$\Leftrightarrow \sin(BDA) + \sin(ADC) = \sin(ABD)$$

$$\begin{aligned} &\Leftrightarrow \sin(120 - ADC) + \sin(ADC) = \sin(60 + ABC) = \sin(60 + ADC) \\ &\Leftrightarrow \frac{\sqrt{3}}{2} \cos(-ADC) + \frac{1}{2} \sin(-ADC) + \sin(ADC) = \frac{\sqrt{3}}{2} \cos(ADC) + \frac{1}{2} \sin(ADC) \end{aligned}$$

The last statement is obviously true.

Therefore,

$$|AB| + |AC| = |OH| \Leftrightarrow |OH| = |AD| \Leftarrow AHOD \text{ is a parallelogram}$$

Since $AH \parallel OD$, we simply need to prove that $AH = OD$.

Note that by reflecting the orthocenter, BCH has the same circumradius as ABC .

$$AH = OD \Leftrightarrow AH = R$$

We let $\angle ABC = x$, so $\angle BAH = 90 + x$, and $\frac{AH}{\sin(30)} = \frac{HB}{\sin(90+x)} = \frac{HB}{\sin(90-x)} = \frac{HB}{\sin(HCB)} = 2R$.

Problem 14 (USAMO 2023/1)

We let D be the base of the perpendicular from A to BC .

$$MQ \cdot MB = MP \cdot MA = MR \cdot MC$$

$$\Leftrightarrow MQ = MR \Leftrightarrow M \text{ is the midpoint of } QR \Leftrightarrow MN \parallel AR \perp BC \Leftrightarrow BN = CN$$

Problem 15 (SL 2006 G2)

(Using directed angles)

Let $AD \cap BC = X$ such that X, L, K are collinear by homothety.

$$\angle XBA = \angle XCD = \angle CQD$$

This gives that BX is tangent to (CDQ) . We let (CDQ) intersect KL again at Y .

Since $\angle DYC = 180 - \angle DQC = \angle APB$, the previous homothety takes Y to P as well.

We can angle chase,

$$\angle QCB = \angle QYC = \angle KPB = \angle QPB$$

Problem 18 (USEMO 2022/4)

All $\angle$ are directed.

We prove that $XY \perp SQ \perp PR$, starting by proving the first part with the

We let $U = AD \cap BC, V = AB \cap CD$.

Claim: $US = UQ, VP = VR$

Note that,

$$\angle BSA = \angle SAB + \angle ABS = \angle BCD + \angle SCD = \angle BCS$$

Therefore, (BSC) is tangent to AD . Similarly, (AQD) is tangent to BC .

$$US^2 = UB \cdot UC = UA \cdot UD = UQ^2$$

Similar for $VP = VR$.

To prove the first part of our main goal, **claim: $SXQY$ is a kite.**

$$\angle BXA = \angle SXQ = \angle QYS = \angle DYC$$

We're done with proving that $\angle QSX = \angle YSQ$.

$$\begin{aligned} \angle QSX &= \angle QSU + \angle USB = \angle UQS + \angle SCB = \angle CQS + \angle SCQ = \angle CSQ \\ &\therefore YS = XS \end{aligned}$$

With that, we have $XY \perp SQ$.

Claim: the angle bisectors of $\angle U$ and $\angle V$ are perpendicular

The angle θ between them is,

$$\begin{aligned} \theta &= \frac{\angle DUC}{2} + \angle DAU + \angle VAD + \frac{\angle CVB}{2} \\ &= 90^\circ - \frac{\angle BCD}{2} - \frac{\angle CDA}{2} + \angle DCB + 90^\circ - \frac{\angle ABC}{2} - \frac{\angle BCD}{2} \\ &= 90^\circ \end{aligned}$$

By first claim, $PR \perp$ angle bisector of $\angle V \perp$ angle bisector of $\angle U \perp SQ$.

33 clubs; 16.5 hours (includes a lot of review time)

Problem 20 (20ELMO4)

Note that the center of (AHP) is the midpoint of HP since $\angle HAP = 90^\circ$. We let A' be the reflection of H over the center of (BHC) .

Claim: A' is also the reflection of A over M .

A is the orthocenter of BHC , and by lemma 1.17 the reflection of the orthocenter over M is the point across from H about the circumcenter.

Claim: A', G, P are collinear.

By a homothety around H of scale 2, since the line through the centers of the circles pass through the midpoint of HG , we have P, G, A' are collinear.

Therefore,

$$\angle HGA' = 90^\circ$$

Claim: $HM \parallel A'G$

We let H' be the reflection of H over M . Note that,

$$\frac{PD'}{HD'} = \frac{AD}{HD} = \frac{A'D'}{H'D'}$$

Therefore, using the intercept theorem, $HH' \parallel PA'$, which is equivalent to what we want to show.

Using the claim, we obviously have,

$$HM \parallel A'G \perp HG$$

Therefore, $\angle MHG = 90^\circ$.

Problem 23 (203TSTST1)

(All angles are directed)

We let B_M be the midpoint of AC and C_M be that of AB .

Claim: $\angle RAC = \angle B_MGA$; $\angle BAS = \angle AGC_M$

Note that $\angle ACR = \angle CGM$, which we are given, gives (CGR) is tangent to AC . Therefore, by Power of a Point,

$$M_B C^2 = MG \cdot MR$$

$$\Leftrightarrow M_B A^2 = MG \cdot MR$$

$$\Leftrightarrow \angle RAC = \angle B_MGA$$

The other angle condition is analogous.

Now we have that,

$$\angle RAS + \angle BAC = \angle BGC$$

$$\Leftrightarrow \angle BAS + \angle RAC = \angle B_MGA + \angle AGC_M$$

This follows from the claim.

Problem 25 (16EGMO4)

Let O_1 be the center of ω_1 , O_2 be that of ω_2 , and O be that of ω .

Let P_2 be the intersection of X_2T_2 and ω , P_1 be that of X_1T_1 and ω . We seek to prove that $P_1 = P_2$.

Let homothety f_1 about T_1 take O to O_1 .

Let homothety f_2 about T_2 take O to O_2 .

Note that,

$$f_1(f_2(O_2)) = O_1;$$

Since $f_1 \circ f_2$ is also a homothety that takes ω_2 to ω_1 , it is a reflection over M the midpoint of O_1O_2 .

Note that $f_1(P_1) = X_1$ and $f_2(X_2) = P_2$. However, we should have, since X_1 is the reflection of X_2 over M ,

$$\begin{aligned} f_1(f_2(X_2)) &= X_1 \\ \Leftrightarrow f_1(P_2) &= f_1(P_1) \end{aligned}$$

Since homotheties are injective, we have $P_1 = P_2$ and we are done.

48 clubs; 20.5 hours